

1 (a) stress = Young modulus $\times$ strain		
$= 1.8 \times 10^{11} \times 8.2 \times 10^{-4}$ or 1.476×10^8	C1	
$= 0.15$ (0.148) GPa	A1	[2]
(b) (i) wavelength = $3 \times 10^8 / 12 \times 10^{12}$	C1	
$= 25 \mu\text{m}$	A1	[2]
(ii) infra-red/IR	B1	[1]
(c) (i) arrow drawn up to the left of 7.5 N force approximately 5° to 40° to west of north	A1	[1]
(ii) 1. correct vector triangle or working to show magnitude of resultant force = 6.6 N allow 6.5 to 6.7 N if scale diagram	M1	[1]
2. magnitude of acceleration = $6.6 / 0.75$ [scale diagram: $(6.5 \text{ to } 6.7) / 0.75$]	C1	
$= 8.8 \text{ ms}^{-2}$ [scale diagram: $8.7 - 8.9 \text{ ms}^{-2}$]	A1	[2]
(iii) 19° [<u>use of scale diagram</u> allow 17° to 21° (a diagram must be seen)]	B1	[1]